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SINTERING FURNACE

Abstract

Provided is a sintering furnace. The sintering furnace includes: a furnace body and a furnace head cover having a feed inlet, the furnace head cover covering a furnace head of the furnace body, the furnace head cover being axially limited relative to the furnace head, the furnace body being rotatable around a central axis relative to the furnace head cover, and the feed inlet being in communication with an interior of the furnace body; a sliding support structure including a first sliding structure configured to support the furnace head cover, the furnace head cover being fixedly connected to the first sliding structure, and the first sliding structure being slidably arranged in a length direction of the furnace body; and a feeding device in communication with the feed inlet through a flexible connection pipe.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of International Application No. PCT/CN2024/093116, filed on May 14, 2024, which claims priorities to Chinese patent applications Nos. 202410423457.7 and 202420728944.X, filed by TINCI MATERIALS (TAIZHOU) CO., LTD. and GUANGZHOU TINCI MATERIALS TECHNOLOGY CO., LTD. on Apr. 9, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the field of sintering equipment technologies, and more specifically, to a sintering furnace.

BACKGROUND

[0003] A sintering process of lithium iron phosphate mostly uses a roller hearth kiln or a pusher kiln as a primary carrier. An application of a large-scale rotary furnace in a high-temperature solid-phase reaction of lithium iron phosphate represents a key industry breakthrough. The rotary furnace offers advantages such as a high degree of automation, large production capacity, and excellent product stability. However, due to a “large size” of the rotary furnace, the rotary furnace has a more complex structure compared with the roller hearth kiln and the pusher kiln, which poses challenges in maintaining operational stability.

SUMMARY

[0004] The present disclosure aims to solve at least one of the technical problems in the related art. To this end, an objective of the present disclosure is to provide a sintering furnace. The sintering furnace is capable of adapting to an expansion or contraction deformation of a furnace body, improving support stability and feeding stability, and achieving relatively high operational stability.

[0005] According to an embodiment of the present disclosure, a sintering furnace is provided. The sintering furnace includes a furnace body, a furnace head cover, a sliding support structure and a feeding device. The furnace head cover has a feed inlet. The furnace head cover is sleeved on a furnace head of the furnace body. The furnace head cover is axially limited relative to the furnace head. The furnace body is rotatable around a central axis relative to the furnace head cover. The feed inlet is in communication with an interior of the furnace body. The sliding support structure includes a first sliding structure configured to support the furnace head cover. The furnace head cover is fixedly connected to the first sliding structure. The first sliding structure is slidably arranged in a length direction of the furnace body. The feeding device is in communication with the feed inlet through a flexible connection pipe.

[0006] With the sintering furnace according to the embodiment of the present disclosure, an end of the furnace body is sealed through a structural engagement between the furnace head cover and the furnace head or through a further sealing structure provided between the furnace head cover and

the furnace head. In addition, through the cooperation of the sliding support structure, the furnace head cover and the flexible connection pipe, the sintering furnace can adapt to an expansion or contraction deformation of the furnace body, in such a manner that the furnace head cover and the furnace body are continuously supported in a stable manner during the expansion or contraction deformation. In this way, a satisfactory axial engagement and a satisfactory circumferential engagement can be achieved between the furnace head cover and the furnace body, which avoids a structural displacement or even a detachment to prevent a seal failure, helping to improve structural stability of the sintering furnace. Further, the flexible connection pipe can deform with a deformation of the furnace body to avoid a structural detachment or damage caused by the deformation of the furnace body, which can otherwise affect material feeding, contributing to improving feeding stability and effectively enhancing operational stability of the sintering furnace. Since the feed inlet is formed at the furnace head cover and the flexible connection pipe is connected to the furnace head cover, a material can enter the furnace body from the furnace head cover at the end of the furnace body, which helps to expand a sintering region within the furnace body while reducing a quantity of holes in the sintering region of the furnace body, contributing to improving a thermal insulation effect of the furnace body.

[0007] Further, the sintering furnace according to the above embodiment of the present disclosure can further provide the following additional technical features.

[0008] According to some embodiments of the present disclosure, the sliding support structure further includes a first fixing structure having a sliding groove extending in the length direction of the furnace body. The sliding groove has a bottom wall and two side walls, the two side walls being opposite to each other in a horizontal direction perpendicular to the length direction of the furnace body. The first sliding structure is slidably arranged at the bottom wall and located between the two side walls.

[0009] According to some embodiments of the present disclosure, a wear-resistant layer is disposed between the first sliding structure and at least one of the bottom wall and the two side walls.

[0010] According to some embodiments of the present disclosure, an extension length of the sliding groove in the length direction of the furnace body is d ; and a maximum displacement of an edge of the furnace head in the length direction of the furnace body during operation is d_0 , where $d \geq d_0$.

[0011] According to some embodiments of the present disclosure, the sliding support structure further includes a first fixing structure. The first sliding structure includes a connection rod, a first connection portion, a second connection portion, and a buffer. The first connection portion is connected to a lower end of the connection rod and slidably engaged with the first fixing structure. The second connection portion is sleeved on the connection rod and is fixedly connected to the furnace head cover. The buffer is sleeved on the connection rod and has an end abutting against the first connection portion and another end abutting against the second connection portion. The buffer is stretchably deformable in a vertical direction.

[0012] According to some embodiments of the present disclosure, an extension length of the flexible connection pipe is L_1 ; in a cold state of the sintering furnace, a straight-line spacing between the feeding device and the feed inlet is L_2 ; and a maximum displacement of the feed inlet in the length direction of the furnace body during operation is L_0 , where $L_1 - L_2 > L_0$.

[0013] According to some embodiments of the present disclosure, the furnace body has a spiral groove spirally extending in a circumferential direction of the furnace body. A reserved space is formed between an end wall of the furnace head cover away from the furnace head and an end portion of the furnace head. The feed inlet is formed at a peripheral wall of the reserved space. The furnace head cover is internally provided with a feed pipe including a first pipe segment and a second pipe segment connected to the first pipe segment, the first pipe segment passing through the reserved space and being connected to the feed inlet, the second pipe segment passing through an end opening of the furnace head and extending to a position above the spiral groove, the first pipe segment extending downwards, and the second pipe segment extending obliquely downwards.

[0014] According to some embodiments of the present disclosure, a packing box is disposed between an inner circumferential surface of the furnace head cover and an outer circumferential surface of the furnace head, the packing box being provided with a plurality of first packings. A gap space is provided between two adjacent first packings of the plurality of first packings. The packing box has a supply hole for supply of a lubricating medium, the supply hole being in communication with at least one of the gap spaces. The sintering furnace further includes a gland connected to the furnace head cover and stopped at a side of the plurality of first packings away from an end wall of the furnace head cover.

[0015] According to some embodiments of the present disclosure, the packing box is further provided with at least one second packing, the at least one second packing being located between the plurality of first packings and the end wall of the furnace head cover, each of the plurality of first packings having a V-shaped axial cross-section, and the second packing having a rectangular axial cross-section.

[0016] According to some embodiments of the present disclosure, the furnace body includes a plurality of furnace segments connected in sequence. A support roller structure is provided at at least one of the furnace head of the furnace body, a furnace tail of the furnace body, and a joint between adjacent furnace segments of the plurality of furnace segments. The support roller structure includes a second fixing structure and at least two support rollers rotatably mounted at the second fixing structure, and a rotation axis of each of the at least two support rollers is parallel to the length direction of the furnace body. The at least two support rollers are supported at a lower side of the furnace body and spaced apart from each other in a horizontal direction. The furnace body is movable relative to each of the at least two support rollers in the length direction.

[0017] According to some embodiments of the present disclosure, an outer circumferential surface of the furnace body is provided with an annular protrusion extending in a circumferential direction. The sintering furnace further includes a limiting device including two limiting rollers and a driving device. The two limiting rollers are disposed at two axial sides of the annular protrusion, respectively, to limit an axial position of the annular protrusion, and each of the two limiting rollers is adapted to roll against the annular protrusion. The driving device is disposed at a side of the limiting device away from the sliding support structure in the length direction of the furnace body. The driving device is in a transmission connection with the furnace body and configured to drive the furnace body to rotate around the central axis.

[0018] According to some embodiments of the present disclosure, a rotation axis of each of the two limiting rollers is away from the annular protrusion in an axial direction of the furnace body and is inclined away from an axis of the furnace body in a radial direction of the furnace body; and in the radial direction of the furnace body, an outer diameter of each of the two limiting rollers gradually increases in a direction away from the axis of the furnace body.

[0019] According to some embodiments of the present disclosure, the sintering furnace further includes a furnace tail cover and a tail support device. The furnace tail cover is sleeved on a furnace tail of the furnace body and rotatably engaged with the furnace tail. The furnace tail cover is fixedly supported at the tail support device, or the tail support device includes a third fixing structure and a second sliding structure, the second sliding structure being configured to support and fix the furnace tail cover, and the second sliding structure being mounted at the third fixing structure and slidable relative to the third fixing structure in the length direction of the furnace body.

[0020] According to some embodiments of the present disclosure, the flexible connection pipe is made of rubber or polypropylene plastic; and each of a feed channel of the feeding device, the furnace head cover, and the furnace body is made of stainless steel or an alloy.

[0021] Additional aspects and advantages of the present disclosure will be provided at least in part in the following description, or will become apparent at least in part from the following description, or can be learned from practicing of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The above and/or additional aspects and advantages of the present disclosure will become more apparent and more understandable from the following description of embodiments taken in conjunction with the accompanying drawings.

[0023] FIG. 1 is a front view of a sintering furnace according to an embodiment of the present disclosure.

[0024] FIG. 2 is a schematic partial structural view of FIG. 1.

[0025] FIG. 3 is a top view of a sintering furnace according to an embodiment of the present disclosure.

[0026] FIG. 4 is a schematic partial structural view of FIG. 3.

[0027] FIG. 5 is a schematic structural view of a sliding support structure in a sintering furnace according to an embodiment of the present disclosure.

[0028] FIG. 6 is a schematic structural view at a flexible connection pipe in a sintering furnace according to an embodiment of the present disclosure.

[0029] FIG. 7 is a partial cross-sectional structural view at a packing box in a sintering furnace according to an embodiment of the present disclosure.

[0030] FIG. 8 is a schematic structural view at a packing box in a sintering furnace according to an embodiment of the present disclosure.

[0031] FIG. 9 is a schematic structural view at a support roller structure in a sintering furnace according to an embodiment of the present disclosure.

[0032] FIG. 10 is a schematic structural view at a limiting device in a sintering furnace according to an embodiment of the present disclosure.

[0033] FIG. 11 is a bottom view of a partial structure of a sintering furnace according to an embodiment of the present disclosure.

[0034] FIG. 12 is an enlarged view at a part indicated by a circle A in FIG. 2.

REFERENCE NUMERALS OF THE ACCOMPANYING DRAWINGS

[0035] sintering furnace **100**; [0036] furnace body **10**; furnace head **11**; furnace segment **12**; first furnace segment **121**; second furnace segment **122**; third furnace segment **123**; furnace tail **13**; annular protrusion **14**; housing **15**; [0037] furnace head cover **20**; feed inlet **21**; feed pipe **22**; first pipe segment **223**; second pipe segment **224**; [0038] sliding support structure **30**; first fixing structure **31**; sliding groove **311**; bottom wall **312**; side wall **313**; [0039] first sliding structure **32**; connection rod **321**; first connection portion **322**; second connection portion **323**; buffer **324**; reinforcing rib **33**; [0040] feeding device **40**; feed outlet **41**; flexible connection pipe **50**; [0041] packing box **60**; second packing **61**; first packing **62**; supply hole **63**; gland **65**; [0042] support roller structure **70**; second fixing structure **71**; support roller **72**; support shaft **73**; side support **74**; [0043] limiting device **80**; limiting roller **81**; screw rod **82**; nut **83**; rotation platform **84**; driving device **85**; motor **86**; reducer **87**; transmission gear **88**; ring gear **89**; [0044] furnace tail cover **90**; tail support device **91**; third fixing structure **96**; second sliding structure **97**.

DETAILED DESCRIPTION

[0045] Embodiments of the present disclosure will be described in detail below with reference to examples thereof as illustrated in the accompanying drawings, throughout which same or similar elements, or elements having same or similar functions, are denoted by same or similar reference numerals. The embodiments described below with reference to the drawings are illustrative only, and are intended to explain, rather than limit, the present disclosure.

[0046] In the description of the present disclosure, it should be understood that, the orientation or the position indicated by terms such as “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “over”, “below”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”,

“bottom”, “inner”, “outer”, “clockwise”, “anti-clockwise”, “axial”, “radial”, and “circumferential” should be construed to refer to the orientation and the position as shown in the drawings, and is only for the convenience of describing the present disclosure and simplifying the description, rather than indicating or implying that the pointed device or element must have a specific orientation, or be constructed and operated in a specific orientation, and therefore cannot be understood as a limitation of the present disclosure.

[0047] In the description of the present disclosure, “first feature” and “second feature” may include one or more of these features; “plurality” means two or more; the first feature “on” or “under” the second feature may mean that the first feature is in direct contact with the second feature, or the first and second features are in indirect contact through another feature between them; the first feature “above” the second feature means that the first feature is directly above or obliquely above the second feature, or simply means that the level of the first feature is higher than that of the second feature.

[0048] A sintering furnace **100** according to the embodiments of the present disclosure is described below with reference to the accompanying drawings.

[0049] Referring to FIG. **1** to FIG. **12**, the sintering furnace **100** according to the embodiments of the present disclosure may include a furnace body **10**, a furnace head cover **20**, a sliding support structure **30**, and a feeding device **40**.

[0050] Specifically, the furnace head cover **20** has a feed inlet **21** and is sleeved on a furnace head **11** of the furnace body **10**. The furnace head cover **20** is axially limited relative to the furnace head **11**. The furnace body **10** is rotatable around a central axis relative to the furnace head cover **20**. The feed inlet **21** is in communication with an interior of the furnace body **10**.

[0051] The sliding support structure **30** includes a first sliding structure **32** configured to support the furnace head cover **20**. The furnace head cover **20** is fixedly connected to the first sliding structure **32**. The first sliding structure **32** is slidably arranged in a length direction (e.g., a front-rear direction as illustrated in FIG. **1** to FIG. **4**) of the furnace body **10**. The feeding device **40** is in communication with the feed inlet **21** through a flexible connection pipe **50**.

[0052] The furnace head cover **20** being axially limited relative to the furnace head **11** means that the furnace head cover **20** is configured to be movable in an axial direction of the furnace body **10** in response to expansion or contraction of the furnace body **10** in length. An end of the furnace body **10** is sealed through a structural engagement between the furnace head cover **20** and the furnace head **11** or through a further sealing structure provided between the furnace head cover **20** and the furnace head **11**. In addition, since the first sliding structure **32** is fixedly connected to the furnace head cover **20**, the first sliding structure **32** fixedly connected to the furnace head cover **20** is slidable in the axial direction when a dimensional deformation occurs in the furnace body **10** due to temperature changes, which allows the furnace head cover **20** to synchronously displace with the furnace head **11** in the axial direction of the furnace body **10**, helping to enable the furnace head cover **20** and the furnace body **10** to be supported in a stable manner during the expansion or contraction of the furnace body **10**. Therefore, reliability of support for the furnace body **10** can be improved. In addition, a satisfactory axial engagement and a satisfactory circumferential engagement can be achieved between the furnace body **10** and the furnace head cover **20**, which avoids a structural displacement or even a detachment to prevent a seal failure that otherwise affects stability of an internal atmosphere of the furnace body **10**, helping to improve structural stability of the sintering furnace **100**. For example, a relative position between the furnace head cover **20** and the furnace head **11** in the axial direction can be fixed using a rigid structure or a flexible structure to reduce relative movements between the furnace head cover **20** and the furnace head **11** in the axial direction, enabling the furnace head cover **20** and the furnace head **11** to move synchronously in the axial direction. However, due to a structural tolerance, a deformation, or other reasons, a slight relative movement between the furnace head cover **20** and the furnace head **11** in the axial direction may still occur. Such scenarios fall within the protection scope of the present

disclosure.

[0053] The furnace body **10** is used for sintering a material and the like, and has an integral cylindrical structure, which may be a single cylindrical body or composed of a plurality of cylindrical segments rigidly connected together. The furnace head cover **20** is capable of supporting the furnace head **11** of the furnace body **10**. In addition, considering that the feed inlet **21** of the furnace head cover **20** is in communication with the interior of the furnace body **10**, the material can be conveyed to the furnace body **10** through the feed inlet **21**. The material from the feeding device **40** enters the feed inlet **21** via the flexible connection pipe **50** and then enters the furnace body **10** for sintering. The feeding device **40** may be a weight-reducing hopper or the like. In some embodiments, the feeding device **40** has a feed outlet **41** in communication with the feed inlet **21** through the flexible connection pipe **50**. The material from the feeding device **40** enters the flexible connection pipe **50** from the feed outlet **41**, and then enters the feed inlet **21**.

[0054] During the sintering of the sintering furnace **100**, a temperature in the furnace body **10** is high, resulting in a relatively large thermal expansion. Therefore, the furnace body **10** is prone to the expansion or contraction deformation, leading to problems such as instability in support. However, with the sliding support structure **30**, the furnace head cover **20**, the furnace body **10**, and the first sliding structure **32** slide together when the furnace body **10** undergoes the expansion or contraction deformation, which helps to adapt to the expansion or contraction deformation of the furnace body **10**. In this way, a possibility of shaking or even collapse of the furnace body **10** can be reduced, improving support stability.

[0055] In addition, the material is conveyed to the furnace body **10** through the flexible connection pipe **50**. The flexible connection pipe **50** is subjected to a deformation (e.g., a stretch deformation, or a bent deformation) in response to the expansion or contraction deformation of the furnace body **10**, to adapt to the expansion or contraction deformation of the furnace body **10** that occurs when the furnace body **10** switches between a cold state and a high-temperature state. In this way, during the deformation of the furnace body **10**, the flexible connection pipe **50** is less likely to detach from the furnace head cover **20**, which results in a relatively stable connection and helps to improve the feeding stability. Further, relatively stable support is provided for the furnace body **10** during the feeding, which helps to further improve the feeding stability. Through an engagement between the sliding support structure **30** and the flexible connection pipe **50**, the sintering furnace **100** can adapt to the expansion or contraction deformation of the furnace body **10**, which is conducive to improving the support stability and the feeding stability, enhancing the operational stability of the sintering furnace **100**. The flexible connection pipe **50** may be made of a deformable material such as polypropylene plastic or rubber.

[0056] For example, in some embodiments, the furnace body **10** has an axial dimension greater than 60 m. During the sintering, a maximum temperature of the furnace body **10** reaches up to 800° C., causing a large thermal expansion of the furnace body **10**. In this case, an axial dimensional change of the furnace body **10** is up to 40 cm. By configuring the furnace head cover **20** and the first sliding structure **32** to allow the furnace body **10** to slide in the length direction, the furnace head cover **20** and the furnace body **10** can be continuously supported during the deformation of the furnace body **10**. In this way, the furnace body **10** is continuously supported by the sliding support structure **30**, achieving relatively strong support stability.

[0057] The feed inlet **21** is provided at the furnace head cover **20**, and the feeding device **40** is in communication with the feed inlet **21** through the flexible connection pipe **50**, in such a manner that the material from the feeding device **40** can enter, via the flexible connection pipe **50**, the furnace body **10** from the furnace head cover **20** at the end of the furnace body **10**, which is conducive to expanding the sintering region within the furnace body **10**. Therefore, no holes need to be formed in the furnace body **10** to convey the material into the furnace body **10**, which reduces the quantity of holes in the sintering region of the furnace body **10** to reduces a heat loss of the furnace body **10**, helping to improve a thermal insulation effect of the furnace body **10**. In this way,

a high-temperature environment is maintained inside the furnace body **10** for high-temperature sintering of the material. Additionally, reducing the quantity of holes makes the furnace body **10** less prone to cracking, helping to enhance strength of the furnace body **10**.

[0058] With the sintering furnace **100** according to the embodiment of the present disclosure, the end of the furnace body **10** is sealed through the structural engagement between the furnace head cover **20** and the furnace head **11** or through the further sealing structure provided between the furnace head cover **20** and the furnace head **11**. In addition, through the cooperation of the sliding support structure **30**, the furnace head cover **20** and the flexible connection pipe **50**, the sintering furnace **100** can adapt to the expansion or contraction deformation of the furnace body **10**, in such a manner that the furnace head cover **20** and the furnace body **10** are continuously supported in a stable manner during the expansion or contraction deformation. In this way, the satisfactory axial engagement and the satisfactory circumferential engagement can be achieved between the furnace head cover **20** and the furnace body **10**, which avoids the structural displacement or even the detachment to prevent the seal failure, helping to improve the structural stability of the sintering furnace **100**. Further, the flexible connection pipe **50** can deform with the deformation of the furnace body **10** to avoid a structural detachment or damage caused by the deformation of the furnace body **10**, which can otherwise affect material feeding, contributing to improving the feeding stability and effectively enhancing the operational stability of the sintering furnace **100**.

Since the feed inlet **21** is provided at the furnace head cover **20** and the flexible connection pipe **50** is connected to the furnace head cover **20**, a material can enter the furnace body **10** from the furnace head cover **20** at the end of the furnace body **10**, which helps to expand the sintering region within the furnace body **10** while reducing the quantity of holes in the sintering region of the furnace body **10**, contributing to improving the thermal insulation effect of the furnace body **10**.

[0059] The first sliding structure **32** can be slidably supported on the ground by mounting wheels at a bottom of the first sliding structure **32** or other means to enable sliding in the length direction of the furnace body **10**. Also, the first sliding structure **32** may be supported by other structures to achieve sliding functionality.

[0060] For example, in some embodiments of the present disclosure, as illustrated in FIG. **1** to FIG. **5**, the sliding support structure **30** further includes a first fixing structure **31** having a sliding groove **311** extending in the length direction of the furnace body **10**. The sliding groove **311** has a bottom wall **312** and two side walls **313**. The two side walls are opposite to each other in a horizontal direction perpendicular to the length direction of the furnace body. The first sliding structure **32** is slidably arranged at the bottom wall **312** and located between the two side walls **313**.

[0061] The first sliding structure **32** is mounted at the first fixing structure **31** and slidable relative to the first fixing structure **31** in the length direction of the furnace body **10**. The bottom wall **312** is capable of supporting the first sliding structure **32** and restricting a displacement of the first sliding structure **32** in an up-down direction, while the two side walls **313** are capable of restricting a displacement of the first sliding structure **32** in the horizontal direction. In this way, the first sliding structure **32** can only slide in the length direction of the furnace body **10**, which reduces a possibility of shaking during sliding of the furnace body **10** through the sliding support structure **30**, improving the support reliability.

[0062] For example, in some specific embodiments, as illustrated in FIG. **5**, the position of the first sliding structure **32** is limited by the sliding groove **311** in the up-down direction and a left-right direction, allowing the first sliding structure **32** to slide only in the front-rear direction. When the furnace body **10** undergoes the expansion or contraction deformation in the front-rear direction, the furnace body **10** is slidable in the front-rear direction to adapt to the deformation, providing relatively reliable support.

[0063] In some embodiments, the first fixing structure **31** includes an I-beam, which is fixedly mounted at a concrete post by a bolt. A sliding rail defining the sliding groove **311** is welded to an upper end face of the I-beam, which is conducive to enhancing connection reliability between the

sliding rail and the I-beam and between the I-beam and the concrete post, improving mounting reliability of the first fixing structure **31**.

[0064] In some embodiments, as illustrated in FIG. 4 and FIG. 5, a reinforcing rib **33** is welded between the sliding rail and the I-beam to enhance mechanical strength of the sliding rail and the I-beam. A quantity and shapes of the reinforcing ribs **33** are not limited in the present disclosure. For example, in some embodiments, four reinforcing ribs **33** are welded in a rectangular arrangement between the sliding rail and the I-beam for ease of processing.

[0065] In some embodiments, as illustrated in FIG. 5, a wear-resistant layer is disposed between the first sliding structure **32** and at least one of the bottom wall **312** and the two side walls **313**, helping to reduce a sliding resistance of the first sliding structure **32** within the sliding groove **311**. In addition, the wear-resistant layer exhibits relatively satisfactory wear resistance, which is conducive to reducing wear and offering relatively satisfactory economic performance.

[0066] For example, in some embodiments, the wear-resistant layer may include a ceramic layer, enabling smoother sliding of the first sliding structure **32** within the sliding groove **311** and providing better wear resistance. For example, in other embodiments, the wear-resistant layer is made of stainless steel and has hardness greater than that of each of the bottom wall **312** and the two side walls **313**, offering relatively satisfactory wear resistance.

[0067] An extension length of the sliding groove **311** in the length direction of the furnace body **10** can be designed based on the length of the furnace body **10**. For example, in some embodiments, as illustrated in FIG. 4, the extension length of the sliding groove **311** in the length direction of the furnace body **10** is d , and a maximum displacement of an edge of the furnace head **11** in the length direction of the furnace body **10** during operation is d_0 , where $d \geq d_0$.

[0068] During operation of the sintering furnace **100**, the maximum displacement of the edge of the furnace head **11** in the length direction of the furnace body **10** is d_0 . For example, the sintering furnace **100** further includes a limiting device **80** connected to the furnace body **10** and spaced apart from the furnace head **11** in the length direction of the furnace body **10**. The limiting device **80** is configured to limit a position of a connected part of the furnace body **10** in the length direction. When the furnace body **10** is in the cold state, a distance between the edge of the furnace head **11** and the limiting device **80** in the length direction of the furnace body **10** is d_1 . During operation, the furnace body **10** undergoes a thermal expansion due to a temperature rise, causing the edge of the furnace head **11** to displace away from the limiting device **80** in the length direction of the furnace body **10**. After an operation condition is stabilized, the distance between the edge of the furnace head **11** and the limiting device **80** in the length direction of the furnace body **10** becomes d_2 , where $d_0 = d_2 - d_1$. A specific value of d_0 depends on factors such as the length of the furnace body **10**, a position of the limiting device **80** in the length direction of the furnace body **10**, a linear expansion coefficient corresponding to a material of the furnace body **10**, and a temperature difference between the cold state and an operation state of the furnace body **10**, and needs to be determined based on actual heat transfer characteristics of the sintering equipment such as the sintering furnace **100** and a specific sintering process of the material.

[0069] To meet a range requirement of the expansion or contraction deformation of the furnace body **10** in the axial direction, the extension length d of the sliding groove **311** in the length direction of the furnace body **10** should be equal to or greater than the maximum displacement d_0 of the edge of the furnace head **11** in the length direction of the furnace body **10**, i.e., $d \geq d_0$. In this way, during operation of the furnace body **10** when the thermal expansion occurs due to the temperature rise, the first sliding structure **32** can be ensured to slide freely within the sliding groove **311** without slipping out of the sliding groove **311**, allowing the first sliding structure **32** to provide stable support for the furnace head cover **20** and the furnace head **11**.

[0070] For example, in some specific embodiments, the furnace body **10** is made of 310S stainless steel. When the furnace body **10** elongates due to the thermal expansion, and the maximum displacement d_0 of the edge of the furnace head **11** in the length direction of the furnace body **10** is

greater than or equal to 20 cm and smaller than or equal to 80 cm, the extension length d of the sliding groove **311** in the length direction of the furnace body **10** is greater than or equal to d_0 , which enables the first sliding structure **32** to slide within the sliding groove **311** at all times without slipping out of the sliding groove **311**, providing relatively broad applicability. For instance, if d_0 is 20 cm, d may be any value greater than or equal to 20 cm, such as 20 cm, 50 cm, or 80 cm. For example, if d_0 is 40 cm, d may be any value greater than or equal to 40 cm, such as 40 cm, 60 cm, or 80 cm. In some specific embodiments, d is greater than or equal to 20 cm and smaller than or equal to 80 cm.

[0071] In some embodiments of the present disclosure, as illustrated in FIG. 5, the sliding support structure **30** further includes a first fixing structure **31**. The first sliding structure **31** includes a connection rod **321**, a first connection portion **322**, a second connection portion **323**, and a buffer **324**. The first connection portion **322** is connected to a lower end of the connection rod **321** and slidably engaged with the first fixing structure **31**. The second connection portion **323** is sleeved on the connection rod **321** and is fixedly connected to the furnace head cover **20**. The buffer **324** is sleeved on the connection rod **321** and has an end abutting against the first connection portion **322** and another end abutting against the second connection portion **323**. The buffer **324** is stretchably deformable in a vertical direction (e.g., in the up-down direction as illustrated in FIG. 5).

[0072] When the furnace body **10** undergoes the expansion or contraction deformation, the first connection portion **322** is slidably engaged with the first fixing structure **31**, enabling the furnace head cover **20** and the furnace body **10** to slide to adapt to the deformation of the furnace body **10** in the axial direction. Since the end of the buffer **324** abuts with the first connection portion **322** and the other end abuts with the second connection portion **323**, the buffer **324** deforms under a force applied by the second connection portion **323** to cushion the deformation of the furnace body **10** in the vertical direction. In addition, the connection rod **321** is capable of constraining the stretch direction of the buffer **324**, which is conducive to meeting a range requirement of the expansion and contraction of the furnace body **10** in the axial direction or the radial direction. The buffer **324** may be a deformable object such as a spring.

[0073] In some specific embodiments, as illustrated in FIG. 5, the first connection portion **322** is a slider that is slidably mounted in the sliding groove **311**, and the second connection portion **323** is a support plate fixedly connected to the furnace head cover **20**. The support plate may be connected to the furnace head cover **20** through welding, riveting, or the like. The support plate is sleeved on the connection rod **321**. The support plate and the connection rod **321** may be connected by means such as riveting to prevent a detachment. The buffer **324** is a spring capable of supporting a load of 1 ton, providing better support for the furnace head cover **20**.

[0074] In some embodiments of the present disclosure, as illustrated in FIG. 6, an extension length of the flexible connection pipe **50** is L_1 . In the cold state of the sintering furnace **100**, a straight-line spacing between the feeding device **40** and the feed inlet **21** is L_2 . A maximum displacement of the feed inlet **21** in the length direction of the furnace body **10** during operation is L_0 , where $L_1 - L_2 > L_0$.

[0075] The extension length of the flexible connection pipe **50** refers to a length of a path extending from an end of the flexible connection pipe **50** to another end of the flexible connection pipe **50** when the flexible connection pipe **50** is undeformed. For example, if the flexible connection pipe **50** extends along an arc, the extension length refers to a length of the arc. Alternatively, if the flexible connection pipe **50** is an elastic and stretchable pipe, the extension length of the flexible connection pipe **50** refers to a maximum length of a path extending from the end of the flexible connection pipe **50** to the other end of the flexible connection pipe **50** within an allowable elastic limit of the flexible connection pipe **50**. The straight-line spacing between the feeding device **40** and the feed inlet **21** may refer to a straight-line spacing between the feed outlet **41** and the feed inlet **21**.

[0076] During operation of the sintering furnace **100**, the maximum displacement of the feed inlet

21 in the length direction of the furnace body **10** is L_0 . For example, the sintering furnace **100** further includes the limiting device **80**. When the furnace body **10** is in the cold state, a distance between the feed inlet **21** and the limiting device **80** in the length direction of the furnace body **10** is L_3 . During operation, the furnace body **10** undergoes the thermal expansion due to the temperature rise, causing the feed inlet **21** to displace away from the limiting device **80** in the length direction of the furnace body **10**. After the operation condition is stabilized, the distance between the feed inlet **21** and the limiting device **80** in the length direction of the furnace body **10** becomes L_4 , where $L_0 = L_4 - L_3$. A specific value of L_0 depends on factors such as the length of the furnace body **10**, the position of the limiting device **80** in the length direction of the furnace body **10**, a linear expansion coefficient corresponding to the material of the furnace body **10**, and the temperature difference between the cold state and the operation state of the furnace body **10**, and needs to be determined based on actual heat transfer characteristics of the sintering equipment and the specific sintering process of the material.

[0077] When the feed inlet **21** of the furnace head cover **11** moves due to the deformation of the furnace body **10**, the flexible connection pipe **50** connecting the feeding device **40** and the feed inlet **21** moves accordingly. Additionally, $L_1 - L_2 > L_0$, which ensures that the length of the flexible connection pipe **50** is sufficient to adapt to a maximum movement range of the feed inlet **21**. In this way, the flexible connection pipe **50** is less likely to detach from the feed inlet **21** during a movement of the feed inlet **21**, improving stability of conveying the material to the furnace body **10** through the flexible connection pipe **50**. Moreover, the sufficient length of the flexible connection pipe **50** makes the flexible connection pipe **50** less likely to be affected by external forces, such as tensile forces from the feeding device **40** and the furnace head cover **11**, which otherwise affects a material conveying function of the flexible connection pipe **50**. Similarly, the feeding device **40** and the furnace head cover **11** are also less likely to be affected by a tensile force from the flexible connection pipe **50** that otherwise affects the operational stability, facilitating smooth sintering. Further, the flexible connection pipe **50** may also be deformable to enable the flexible connection pipe **50** to adapt to a larger movement range of the feed inlet **21**, achieving broader applicability.

[0078] For example, if L_0 is 20 cm, $L_1 - L_2$ may be any value greater than or equal to 20 cm, such as 21 cm, 25 cm, or 30 cm. For example, if L_0 is 80 cm, $L_1 - L_2$ may be any value greater than or equal to 80 cm, such as 81 cm, 85 cm, or 90 cm.

[0079] In some embodiments of the present disclosure, as illustrated in FIG. 6, the furnace body **10** has a spiral groove spirally extending in a circumferential direction of the furnace body **10**. A reserved space is provided between an end wall of the furnace head cover **20** away from the furnace head **11** and an end portion of the furnace head **11**, and the reserved space is formed by the end wall of the furnace head cover **20** away from the furnace head **11** and the end portion of the furnace head **11**. The feed inlet **21** is provided at a peripheral wall of the reserved space. The furnace head cover **20** is internally provided with a feed pipe **22** including a first pipe segment **223** and a second pipe segment **224** connected to the first pipe segment **223**. The first pipe segment **223** passes through the reserved space and is connected to the feed inlet **21**. The second pipe segment **224** passes through an end opening of the furnace head **11** and extends to a position above the spiral groove. The first pipe segment **223** extends downwards. The second pipe segment **224** extends obliquely downwards.

[0080] The spiral groove is capable of guiding the material. For example, in some embodiments, the spiral groove is defined by a helical blade protruding from an inner wall of the furnace body **10**. The helical blade extends spirally in the axial direction of the furnace body **10**. When the furnace body **10** rotates, the material inside the furnace body **10** is turned over and redistributed under guidance of the helical blade, which pushes the material forwards. In this way, uniformity of distribution of the material is improved, which facilitates the sintering of the material.

[0081] The furnace head cover **20** has an end (e.g., a rear end as illustrated in FIG. 6) formed as an opening and connected to the end of the furnace head **11** and another end (e.g., a front end as

illustrated in FIG. 6) formed as an end wall forming the reserved space with the end of the furnace head **11**. Such a configuration allows the first pipe segment **223** to extend downwards to pass through the reserved space and the second pipe segment **224** to extend obliquely downwards to pass through the end opening of the furnace head **11**, which can reduce positional conflicts between the furnace body **10** and the feed pipe **22** during a rotation of the furnace body **10**, improving the feeding stability.

[0082] The first pipe segment **223** may extend vertically downwards or obliquely downwards. The feed pipe **22** may be disposed in the furnace head cover **20** using methods such as welding. For example, in some embodiments, the feed pipe **22** is welded to the furnace head cover **20** using argon arc welding.

[0083] The material flows from the feed inlet **21** of the furnace head cover **20** to the first pipe segment **223** and moves towards the second pipe segment **224** under gravity, without using an external feeding tool, which facilitates guidance of the material to the second pipe segment **224** and subsequent flowing of the material into the spiral groove, ensuring stable conveying of the material.

[0084] In some embodiments, as illustrated in FIG. 6, an angle between the second pipe segment **224** and the horizontal direction is greater than an angle of repose of the material, in such a manner that the material in the second pipe segment **224** is likely to flow downwards towards the spiral groove under gravity. Consequently, a possibility of blockages of the material in the feed pipe **22** is reduced, which is beneficial to realize stable feeding of the material. For example, the material is lithium iron phosphate, and the angle between the second pipe segment **224** and the horizontal direction is greater than the angle of repose of lithium iron phosphate.

[0085] Additionally, since the sintering of the material such as a positive electrode material for lithium-ion batteries requires a protective gas atmosphere, high requirements are placed on sealing performance of the sintering equipment. Therefore, maintaining gas tightness is also a significant challenge for an application of a large-scale sintering furnace in the sintering such as a high-temperature solid-phase reaction of lithium iron phosphate. In particular, while ensuring reliable sealing, the sealing structure needs to be structurally well adapted to a rotation of a movement part (such as the furnace body) of the sintering furnace during operation, a radial deformation and an axial deformation of the furnace body caused by switching between the cold state and a hot state, a shape error (e.g., ovality or eccentricity) and bending of a geometric centerline of the furnace body during manufacturing, mounting, or transportation.

[0086] Based on this, the present disclosure provides the following improvements to meet a sealing requirement of the sintering. In some embodiments of the present disclosure, as illustrated in FIG. 1 to FIG. 4, FIG. 7, and FIG. 8, a packing box **60** is disposed between an inner circumferential surface of the furnace head cover **20** and an outer circumferential surface of the furnace head **11**. The packing box **60** is provided with a plurality of first packings **62**. The first packing **62** is capable of sealing a connection gap between the furnace head cover **20** and the furnace head **11** to improve the sealing performance between the furnace head cover **20** and the furnace head **11**, which helps to maintain a protective gas atmosphere such as an inert gas atmosphere inside the sintering furnace **100**, preventing an excessive oxygen level from affecting roasting of the material such as lithium iron phosphate. For example, the excessive oxygen level can lead to more lithium iron phosphate oxidizing into byproducts such as ferric salts. In addition, a possibility of the material in the sintering furnace **100** flowing towards an ambient environment or dust or the like from the ambient environment flowing towards the sintering furnace **100** can be reduced to improve the sealing performance between the furnace head cover **20** and the furnace head **11**, improving a roasting indicator and a safety and environmental indicator of the sintering furnace **100**. The first packing **62** may be made of ceramic fiber, rubber, etc.

[0087] In some embodiments, as illustrated in FIG. 7, a gap space is provided between two adjacent first packings **62** of the plurality of first packings **62**, and the packing box **60** has a supply hole **63** for supplying a lubricating medium (e.g., grease). The supply hole **63** is in communication

with at least one gap space. With the supply hole **63**, the lubricating medium can be introduced into the at least one gap space, which helps to reduce corrosion and wear on a contact surface between the furnace body **10** and the packing, reducing a possibility of rust formation on the furnace body **10**, and extending a service life of the furnace body **10**. For example, in some embodiments, as illustrated in FIG. 7, the gap space and the supply hole **63** at the packing box **60** are opposite to each other in the radial direction and in communication with each other.

[0088] In some embodiments, as illustrated in FIG. 7 and FIG. 8, the sintering furnace **100** further includes a gland **65**. The gland **65** is connected to the furnace head cover **20** and stopped at a side of the plurality of first packings **62** away from the furnace head cover **20**. In this way, fixation of a position of the first packing **62** can be facilitated, enhancing mounting stability and improving sealing effectiveness and sealing longevity of the first packing **62**. For example, in some embodiments, as illustrated in FIG. 7, the gland **65** is stopped at a rear side of the first packing **62** and the packing box **60**, to fix positions of the first packing **62** and the packing box **60** in the front-rear direction.

[0089] In some embodiments, as illustrated in FIG. 7, the packing box **60** is further provided with at least one second packing **61**. The at least one second packing **61** is located between the plurality of first packings **62** and the end wall of the furnace head cover **20**. Each of the plurality of first packings **62** has a V-shaped axial cross-section. The second packing **61** has a rectangular axial cross-section. The axial cross-section of the packing refers to a cross-section formed by the packing after the packing is intersected by a plane containing an axis of the furnace body **10**. The packing can reduce a possibility of the sealing failure during the deformation of the furnace body **10**. In addition, cooperation between the first packing **62** and the second packing **61** in different shapes is conducive to improving the sealing performance.

[0090] For example, in some embodiments, as illustrated in FIG. 7, two second packings **61** and six first packings **62** are arranged in sequence in the axial direction between the packing box **60** and the furnace head **11**. The two second packings **61** form one group, and every three adjacent first packings **62** of the six first packings **62** form another group, creating three groups of packings. Cooling water may be introduced between two adjacent first packings **62** in one group to cool the packings and reduce a possibility of damage to the packings due to an excessive friction temperature. An inert gas such as nitrogen may be introduced between the second packing **61** and the first packing **62** that are adjacent each other. The cooling water and the inert gas are capable of compressing the first packings **62**, causing the first packings **62** to expand in the radial direction. In this way, a connection among the furnace head cover **20**, the packing, and the furnace head **11** becomes tighter, improving the sealing performance.

[0091] The furnace head **11** is rotatable relative to the furnace head cover **20**. With the packing box **60**, effective sealing can be achieved between a fixed part (i.e., the furnace head cover **20**) and a rotatable part (i.e., the furnace head **11**) of the sintering furnace **100** to ensure the sealing performance. Additionally, each of the first packing **62** and second packing **61** may be made of rubber or ceramic fiber, which provides each of the first packing **62** and second packing **61** with a predetermined degree of deformation allowance. When the furnace body **10** undergoes the expansion or the deformation in the radial direction due to factors such as a temperature difference during rotation and compresses the packing, the packing can deform in response to the deformation of the furnace body **10** to be in continuous sealing contact with the furnace body **10**, which further improves the sealing performance. Moreover, the packing can deform to adapt to the shape error (e.g., ovality or eccentricity) and the bending of the geometric centerline of the furnace body **10** during manufacturing, mounting, or transportation. In addition, when the furnace body **10** undergoes a relatively large dimensional change in the length direction of the furnace body **10**, e.g., the thermal expansion, since the furnace head cover **20** is stably supported by the sliding support structure **30**, the furnace head cover **20** can slide in the length direction of the furnace body **10** in accordance with the deformation of the furnace body **10**. Therefore, during the deformation of the

furnace body **10**, the furnace head cover **20**, the packing box **60**, and the furnace head **11** remain tightly connected to ensure the sealing performance, maintaining a stable inert gas atmosphere inside the furnace body **10**.

[0092] It should be noted that the relative position between the furnace head cover **20** and the furnace head **11** in the axial direction may change due to the deformation of the furnace body **10**. However, in the present disclosure, the packing box **60** always ensures a sealed connection between the furnace head cover **20** and the packing box **60**, in such a manner that the relative position between the furnace head cover **20** and the furnace head **11** in the axial direction changes relatively little or even remains substantially unchanged, which is conducive to supporting the furnace head cover **20** by the first sliding structure **32** and therefore supporting the furnace body **10**, improving the reliability of the support for the furnace body **10**.

[0093] In some embodiments of the present disclosure, as illustrated in FIG. **1** to FIG. **4** and FIG. **9**, the furnace body **10** includes a plurality of furnace segments **12** connected in sequence. A support roller structure **70** is provided at at least one of the furnace head **11** of the furnace body **10**, a furnace tail **13** of the furnace body **10**, and a joint between adjacent furnace segments **12** of the plurality of furnace segments **12**. The support roller structure **70** includes a second fixing structure **71** and at least two support rollers **72** rotatably mounted at the second fixing structure **71**, and a rotation axis of each of the at least two support rollers **72** is parallel to the length direction of the furnace body **10**. The at least two support rollers **72** are supported at a lower side of the furnace body **10** and spaced apart from each other in a horizontal direction. The furnace body **10** is movable relative to each of the at least two support rollers **72** in the length direction.

[0094] The plurality of furnace segments **12** are capable of performing different processing tasks on the material. Since the furnace body **10** is relatively long in the axial direction of the furnace body **10**, at least one support roller structure **70** can support the furnace body **10**. In addition, the support roller structure **70** cooperates with the sliding support structure **30** at the end to support the entire furnace body **10** and the furnace head cover **20**, which is more conducive to enhancing the operational stability of the sintering furnace **100**. In some embodiments of the present disclosure, at least four support roller structures **70** are used to support the furnace body **10**, providing more stable support for the sintering furnace **100**. The support roller **72** is rotatable relative to the second fixing structure **71**, and the furnace body **10** is movable relative to the support roller **72** in the length direction, which is conducive to satisfying the rotation of the furnace body **10** and the expansion or contraction deformation of the furnace body **10** in the length direction. By supporting the furnace body **10** with the support roller **72**, and with the cooperation of the support roller **72** and the sliding support structure **30**, shaking of the furnace body **10** in the radial direction is reduced while adapting to the expansion or contraction deformation of the furnace body **10**, which helps to improve sealing reliability between the furnace head **11** and the furnace head cover **20**. The support roller **72** may be made of Cr2-series cold-rolled steel or the like.

[0095] For example, in some embodiments, as illustrated in FIG. **1** to FIG. **4**, the furnace body **10** includes a first furnace segment **121**, a second furnace segment **122**, and a third furnace segment **123** that are arranged from front to rear. The first furnace segment **121** is used for heating the material. The second furnace segment **122** is used for heat preservation of the material. The third furnace segment **123** is used for cooling the material. Four support roller structures **70** are used to support the furnace head **11**, a joint between the first furnace segment **121** and the second furnace segment **122**, a joint between the second furnace segment **122** and the third furnace segment **123**, and the furnace tail **13**, respectively, providing relatively reliable support for the furnace body **10**. In addition, the furnace segment **12** is externally provided with a housing **15**. The housing **15** is fixed to the concrete post. The furnace segment **12** rotates in the housing **15** around an axis relative to the housing **15**, providing relatively high support stability. Further, a length of each of the support rollers **72** of two front support roller structures **70** in the front-rear direction is greater than that of each of the support rollers **72** of two rear support roller structures **70**, which is conducive to

adapting to a forward expansion or contraction deformation of the heating segment and the heat preservation segment of the furnace body **10** that are at relatively high temperatures. In this way, a risk of the furnace body **10** detaching from the support roller **72** can be reduced, ensuring stable operation.

[0096] In some embodiments, an extension length of the support roller **72** in the length direction of the furnace body **10** is greater than or equal to 20 cm and smaller than or equal to 80 cm, allowing the furnace body **10** to slide, on the support roller **72** relative to the second fixing structure **71**, freely in the length direction during the deformation, which helps to meet the range requirement of the expansion or contraction deformation of the furnace body **10** in the axial direction. For example, the extension length of the support roller **72** in the length direction of the furnace body **10** may be 20 cm, 60 cm, 80 cm, or the like.

[0097] In some specific embodiments, as illustrated in FIG. **9**, the support roller structure **70** further includes two support shafts **73** and four side supports **74**. The two support shafts **73** are spaced apart in the left-right direction and correspond to the two support rollers **72**, respectively. The support roller **72** is sleeved on the support shaft **73**. Among the four side supports **74**, two side supports **74** are respectively arranged corresponding to a front end and a rear end of the left support shaft **73** in the front-rear direction, and the other two side supports **74** are respectively arranged corresponding to a front end and a rear end of the right support shaft **73** in the front-rear direction. The support shaft **73** is fixedly mounted at the side support **74** by one or more means such as a bolt and a snap. The side support **74** is fixedly mounted at the second fixing structure **71** by one or more means such as a bolt and a snap. The second fixing structure **71** is an I-beam truss made of stainless steel and fixedly mounted at the concrete post by a bolt, which is conducive to enhancing rigidity of the support roller structure **70** and the support stability for the furnace body **10**. Further, in one support roller structure **70**, extension lines of tangents at contact points between the two support rollers **72** and the furnace body **10** form an angle of 120° (as indicated by α in FIG. **9**), providing more stable support for the furnace body **10**. In some embodiments, the support roller structure **70** is capable of supporting the furnace body **10** weighing 38 tons.

[0098] In some embodiments of the present disclosure, as illustrated in FIG. **1** to FIG. **4**, FIG. **10**, and FIG. **11**, an outer circumferential surface of the furnace body **10** is provided with an annular protrusion **14** extending in a circumferential direction, and the sintering furnace **100** further includes a limiting device **80** and a driving device **85**.

[0099] The limiting device **80** includes two limiting rollers **81**. The two limiting rollers **81** are disposed at two axial sides of the annular protrusion **14**, respectively, to limit an axial position of the annular protrusion **14**. Each of the two limiting rollers **81** is adapted to roll against the annular protrusion **14**. The driving device **85** is disposed at a side of the limiting device **80** away from the sliding support structure **30** in the length direction of the furnace body **10**. The driving device **85** is in a transmission connection with the furnace body **10** and configured to drive the furnace body **10** to rotate around the central axis.

[0100] When the furnace body **10** rotates around the axis of the furnace body **10**, the annular protrusion **14** rotates accordingly. Further, the two limiting rollers **81** are configured to clamp the annular protrusion **14** from two axial sides of the annular protrusion **14** and roll against the annular protrusion **14**, to limit an axial position of the annular protrusion **14**. Consequently, a displacement of the furnace body **10** at the annular protrusion **14** in the axial direction can be restricted to reduce an adverse effect of an axial movement of the furnace body **10** generated when the driving device **85** drives the furnace body **10** to rotate, which is conducive to improving driving stability of the driving device **85** for the furnace body **10**. The annular protrusion **14** may be a flange or the like.

[0101] For example, in some specific embodiments, as illustrated in FIG. **11**, the front limiting roller **81** rotates in a counterclockwise direction as illustrated in FIG. **11**, while the rear limiting roller **81** rotates in a clockwise direction as illustrated in FIG. **11**. The annular protrusion **14** located between the two limiting rollers **81** rotates around an axis extending in the front-rear direction.

[0102] Arranging the driving device **85** at the side of the limiting device **80** away from the sliding support structure **30** helps to reduce the deformation of the furnace body **10** at a position of the driving device **85** and improve the driving stability of the driving device **85**. For example, in some embodiments, as illustrated in FIG. **1** to FIG. **4**, a position of the annular protrusion **14** in a three-dimensional space is substantially fixed by the limiting device **80**. When the furnace body **10** expands and elongates, the furnace body **10** extends forwards with the annular protrusion **14** as an origin, and does not extend backwards under a positional limitation of the limiting device **80**. In this way, the driving device **85** can maintain stable transmission cooperation with the furnace body **10** at a rear side of the limiting device **80**.

[0103] In some embodiments, as illustrated in FIG. **2** and FIG. **4**, the furnace body **10** is provided with a ring gear **89** fixedly mounted at the outer circumferential surface of the furnace body **10**, and the driving device **85** includes a motor **86**, a reducer **87**, and a transmission gear **88**. The transmission gear **88** is engaged with the ring gear **89**. The motor **86** is configured to drive the transmission gear **88** to rotate, which in turn drives the ring gear **89** to rotate, thereby driving the furnace body **10** to rotate. The reducer **87** is configured to increase output torque of the motor **86**, providing relatively satisfactory driving performance. With the limiting device **80**, the transmission gear **88** can remain in a position engaged with the ring gear **89** without moving, improving the driving stability.

[0104] In some embodiments, the support roller structure **70** supports the annular protrusion **14** and therefore supports the furnace body **10**. In this way, a localized force acting on the furnace body **10** can be reduced, helping to extend the service life of the furnace body **10**.

[0105] In some embodiments, as illustrated in FIG. **2**, FIG. **10**, and FIG. **12**, a rotation axis of each of the two limiting rollers **81** is away from the annular protrusion **14** in an axial direction of the furnace body **10** and is inclined away from an axis of the furnace body **10** in a radial direction of the furnace body **10**, and in the radial direction of the furnace body **10**, an outer diameter of each of the two limiting rollers **81** gradually increases in a direction away from the axis of the furnace body **10**. That is, the limiting roller **81** is in a frustum shape and obliquely mounted. Such a design can allow a force exerted by the annular protrusion **14** on the limiting roller **81** to not be perpendicular to an axis of the limiting roller **81** when the limiting roller **81** rolls against the annular protrusion **14**, which reduces a magnitude of a force exerted on the limiting roller **81** in a direction perpendicular to the axis of the limiting roller **81**. In this way, a risk of damage to the limiting roller **81** is lowered, extending a service life of the limiting roller **81**.

[0106] For example, in some embodiments, as illustrated in FIG. **2** and FIG. **12**, the limiting roller **81** is disposed at a lower side of the furnace body **10**, and has a bottom radial outer diameter greater than a top radial outer diameter of the limiting roller **81** to form the frustum shape. The rotation axis of the front limiting roller **81** inclines forwards and downwards, while the rotation axis of the rear limiting roller **81** inclines backwards and downwards.

[0107] In some embodiments, as illustrated in FIG. **12**, the limiting device **80** further includes two rotation platforms **84**, two screw rods **82**, and several nuts **83**. The two rotation platforms **84** are disposed at two axial sides of the annular protrusion **14**, respectively. The two limiting rollers **81** are rotatably disposed at the two rotation platforms **84**, respectively. Each of the two rotation platforms **84** is provided with a positioning boss at a same side of the furnace body **10** in the left-right direction. The positioning boss has a through hole. The screw rod **82** passes through the through holes of the positioning bosses of the two rotation platforms **84**. The nut **83** is disposed at the screw rod **82** and is located at least at a side of the positioning boss away from the annular protrusion **14**. That is, the front nut **83** is disposed at a front side of the front positioning boss, while the rear nut **83** is disposed at a rear side of the rear positioning boss, or each positioning boss is provided with the nut **83** at each of two sides of the positioning boss and tightly compressed by the nuts **83** at the two sides of the positioning boss. The nut **83** is thread-connected to the screw rod **82** to limit the position of the limiting roller **81**, which reduces a possibility of the limiting roller **81** to

move away from the annular protrusion **14** in the axial direction, improving reliability of the limiting device **80** in limiting the position of the annular protrusion **14**.

[0108] In some embodiments, as illustrated in FIG. **1** to FIG. **4**, the sintering furnace **100** further includes a furnace tail cover **90** and a tail support device **91**. The furnace tail cover **90** is sleeved on the furnace tail **13** of the furnace body **10** and is rotatably engaged with the furnace tail **13**. The furnace tail cover **90** is capable of supporting the furnace tail **13** to support the furnace body **10**.

[0109] In some embodiments, the furnace tail cover **90** is fixedly supported at the tail support device **91**, which helps to improve mounting stability of the furnace tail cover **90** and the reliability of the support for the furnace body **10**.

[0110] In some embodiments, as illustrated in FIG. **1** to FIG. **4**, the tail support device **91** includes a third fixing structure **96** and a second sliding structure **97**. The second sliding structure **97** is configured to support and fix the furnace tail cover **90**. In addition, the second sliding structure **97** is mounted at the third fixing structure **96** and slidable relative to the third fixing structure **96** in the length direction of the furnace body **10**. When the furnace body **10** undergoes the expansion or contraction deformation, the furnace tail cover **90**, the furnace body **10**, and the second sliding structure **97** can slide together relative to the third fixing structure **96**, which is conducive to adapting to the expansion or contraction deformation of the furnace body **10**, improving the support stability.

[0111] In the furnace body **10**, a temperature near the furnace tail **13** is lower than that near the furnace head **11**. That is, the deformation of a part of the furnace body **10** between the furnace tail **13** and the limiting device **80** is smaller than that of a part of the furnace body **10** between the furnace head **11** and the limiting device **80**. In some embodiments, a slidable distance of the second sliding structure **97** is smaller than that of the first sliding structure **32**, which helps to adapt to the expansion or contraction deformation of the furnace body **10** while saving costs.

[0112] In some embodiments, as illustrated in FIG. **2** and FIG. **4**, the third fixing structure **96** has a recess extending in the length direction of the furnace body **10**, and the second sliding structure **97** is disposed in the recess of the third fixing structure **96**. An operation principle of the third fixing structure **96** and the second sliding structure **97** is the same as that of the first fixing structure **31** and the second sliding structure **32** in the sliding support structure **30**, and thus details thereof will be omitted here.

[0113] In some embodiments, the packing box **60** is disposed between the furnace tail **13** and the furnace tail cover **90**, and at least one second packing **61** and a plurality of first packings **62** are disposed between the packing box **60** and the furnace tail **13**. In this way, an improvement of sealing performance between the furnace tail **13** and the furnace tail cover **90** is facilitated.

[0114] In some embodiments of the present disclosure, the flexible connection pipe **50** is made of a material such as rubber or polypropylene plastic. In this way, a deformation of the flexible connection pipe **50** is facilitated to improve the feeding stability.

[0115] In some embodiments, each of a feed channel of the feeding device **40**, the furnace head cover **20**, and the furnace body **10** is made of an alloy. In this way, strength of the feed channel, the furnace head cover **20**, and the furnace body **10** can be enhanced to extend a service life of each of the feed channel, the furnace head cover **20**, and the furnace body **10**, which facilitates normal operation of the sintering.

[0116] In some embodiments, each of the feed channel, the furnace head cover **20**, and the furnace body **10** is made of stainless steel. Since stainless steel contains no copper-zinc materials, a possibility of introducing magnetic substances during the high-temperature sintering can be reduced while ensuring the strength of the feed channel, the furnace head cover **20**, and the furnace body **10**, which is beneficial for magnetic substance control. Some materials have extremely strict requirements for magnetic substance control. For example, lithium iron phosphate requires a magnetic substance level in a final product to be smaller than 0.8 ppm. Consequently, using the feed channel, the furnace head cover **20**, and the furnace body **10** made of stainless steel helps to

meet the requirements for magnetic substance control of materials.

[0117] In some embodiments, each of the feed channel, the furnace body **10**, the furnace head cover **20**, and the furnace tail cover **90** is made of 310S stainless steel or an Inconel 601/625 alloy, which can withstand a high temperature ranging from 1,000° C. to 1,200° C. In addition, 310S stainless steel or the Inconel 601/625 alloy provides satisfactory rust-resistant and corrosion-resistant properties and relatively satisfactory intergranular corrosion resistance, which is beneficial to effectively reducing a possibility of introducing the magnetic substances during the high-temperature sintering.

[0118] Other components and operations of the sintering furnace **100** according to the embodiments of the present disclosure are well known to those skilled in the art, and thus details thereof will be omitted here.

[0119] In the description of the present disclosure, it should be noted that, unless otherwise clearly specified and limited, terms such as “install”, “connect”, and “connect to” should be understood in a broad sense. For example, it may be a fixed connection or a detachable connection or connection as one piece; mechanical connection or electrical connection; direct connection or indirect connection through an intermediate; internal communication of two components. For those skilled in the art, the specific meaning of the above-mentioned terms in the present disclosure can be understood according to specific circumstances.

[0120] Reference throughout this specification to “an embodiment”, “a specific embodiment”, or “an example” means that a particular feature, structure, material, or characteristic described in connection with the embodiment or example is included in at least one embodiment or example of the present disclosure. The appearances of the above phrases in various places throughout this specification are not necessarily referring to the same embodiment or example. Further, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments or examples.

[0121] Although embodiments of the present disclosure have been illustrated and described, it is conceivable for those skilled in the art that various changes, modifications, replacements, and variations can be made to these embodiments without departing from the principles and spirit of the present disclosure. The scope of the present disclosure shall be defined by the claims as appended and their equivalents.

Claims

1. A sintering furnace, comprising: a furnace body; a furnace head cover having a feed inlet, the furnace head cover being sleeved on a furnace head of the furnace body, the furnace head cover being axially limited relative to the furnace head, the furnace body being rotatable around a central axis relative to the furnace head cover, and the feed inlet being in communication with an interior of the furnace body; a sliding support structure comprising a first sliding structure configured to support the furnace head cover, the furnace head cover being fixedly connected to the first sliding structure, and the first sliding structure being slidably arranged in a length direction of the furnace body; and a feeding device in communication with the feed inlet through a flexible connection pipe.

2. The sintering furnace according to claim 1, wherein the sliding support structure further comprises: a first fixing structure having a sliding groove extending in the length direction of the furnace body, the sliding groove having a bottom wall and two side walls, the two side walls being opposite to each other in a horizontal direction perpendicular to the length direction of the furnace body, and the first sliding structure being slidably arranged at the bottom wall and located between the two side walls.

3. The sintering furnace according to claim 2, wherein a wear-resistant layer is disposed between the first sliding structure and at least one of the bottom wall and the two side walls.

4. The sintering furnace according to claim 2, wherein: an extension length of the sliding groove in the length direction of the furnace body is d ; and a maximum displacement of an edge of the furnace head in the length direction of the furnace body during operation is d_0 , where $d \geq d_0$.
5. The sintering furnace according to claim 1, wherein: the sliding support structure further comprises a first fixing structure; and the first sliding structure comprises a connection rod, a first connection portion, a second connection portion, and a buffer, the first connection portion being connected to a lower end of the connection rod and slidably engaged with the first fixing structure, the second connection portion being sleeved on the connection rod and fixedly connected to the furnace head cover, the buffer being sleeved on the connection rod and stretchably deformable in a vertical direction, and the buffer having an end abutting against the first connection portion and another end abutting against the second connection portion.
6. The sintering furnace according to claim 1, wherein: an extension length of the flexible connection pipe is L_1 ; in a cold state of the sintering furnace, a straight-line spacing between the feeding device and the feed inlet is L_2 ; and a maximum displacement of the feed inlet in the length direction of the furnace body during operation is L_0 , where $L_1 - L_2 > L_0$.
7. The sintering furnace according to claim 1, wherein: the furnace body has a spiral groove spirally extending in a circumferential direction of the furnace body; a reserved space is provided between an end wall of the furnace head cover away from the furnace head and an end portion of the furnace head; the feed inlet is provided at a peripheral wall of the reserved space; and the furnace head cover is internally provided with a feed pipe comprising a first pipe segment and a second pipe segment connected to the first pipe segment, the first pipe segment passing through the reserved space and being connected to the feed inlet, the second pipe segment passing through an end opening of the furnace head and extending to a position above the spiral groove, the first pipe segment extending downwards, and the second pipe segment extending obliquely downwards.
8. The sintering furnace according to claim 1, wherein: a packing box is disposed between an inner circumferential surface of the furnace head cover and an outer circumferential surface of the furnace head, the packing box being provided with a plurality of first packings; a gap space is provided between two adjacent first packings of the plurality of first packings; the packing box has a supply hole for supply of a lubricating medium, the supply hole being in communication with at least one of the gap spaces; and the sintering furnace further comprises a gland connected to the furnace head cover and stopped at a side of the plurality of first packings away from an end wall of the furnace head cover.
9. The sintering furnace according to claim 8, wherein the packing box is further provided with at least one second packing located between the plurality of first packings and the end wall of the furnace head cover, each of the plurality of first packings having a V-shaped axial cross-section, and the second packing having a rectangular axial cross-section.
10. The sintering furnace according to claim 1, wherein: the furnace body comprises a plurality of furnace segments connected in sequence; a support roller structure is provided at at least one of the furnace head of the furnace body, a furnace tail of the furnace body, and a joint between adjacent furnace segments of the plurality of furnace segments; the support roller structure comprises a second fixing structure and at least two support rollers rotatably mounted at the second fixing structure, a rotation axis of each of the at least two support rollers being parallel to the length direction of the furnace body, the at least two support rollers being supported at a lower side of the furnace body and spaced apart from each other in a horizontal direction, and the furnace body being movable relative to each of the at least two support rollers in the length direction.
11. The sintering furnace according to claim 1, wherein an outer circumferential surface of the furnace body is provided with an annular protrusion extending in a circumferential direction, and wherein the sintering furnace further comprises: a limiting device comprising two limiting rollers, the two limiting rollers being disposed at two axial sides of the annular protrusion, respectively, to limit an axial position of the annular protrusion, and each of the two limiting rollers being adapted

to roll against the annular protrusion; and a driving device disposed at a side of the limiting device away from the sliding support structure in the length direction of the furnace body, the driving device being in a transmission connection with the furnace body and configured to drive the furnace body to rotate around the central axis.

12. The sintering furnace according to claim 11, wherein: a rotation axis of each of the two limiting rollers is away from the annular protrusion in an axial direction of the furnace body and is inclined away from an axis of the furnace body in a radial direction of the furnace body; and in the radial direction of the furnace body, an outer diameter of each of the two limiting rollers gradually increases in a direction away from the axis of the furnace body.

13. The sintering furnace according to claim 11, further comprising: a furnace tail cover and a tail support device, the furnace tail cover being sleeved on a furnace tail of the furnace body and rotatably engaged with the furnace tail, wherein: the furnace tail cover is fixedly supported at the tail support device, or the tail support device comprises a third fixing structure and a second sliding structure, the second sliding structure being configured to support and fix the furnace tail cover, and the second sliding structure being mounted at the third fixing structure and slidable relative to the third fixing structure in the length direction of the furnace body.

14. The sintering furnace according to claim 1, wherein: the flexible connection pipe is made of rubber or polypropylene plastic; and each of a feed channel of the feeding device, the furnace head cover, and the furnace body is made of stainless steel or an alloy.
